CLEARENTRY AI

Technical Design Document

AI-assisted customs document extraction, validation and review workflow

Repository: https://github.com/muralivijai64/ClearEntry-AI

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1. Executive Summary

ClearEntry AI is a demonstration workflow for customs-broker teams handling import shipment documentation. It accepts an invoice, packing list and Bill of Lading (BOL), extracts selected shipment fields with an LLM, compares values across documents, and routes uncertain or inconsistent records to a human reviewer. A manager can approve a shipment only after the required checks are clear and review obligations are complete. The approved shipment can then be exported as a structured Excel workbook.

The system is decision-support software. It does not file a Bill of Entry, connect to a customs portal, make legal determinations, or independently assign/classify HSN codes or calculate duties.

2. Scope and Design Goals

In scope

- JWT-based login with executive and manager roles.
- Shipment creation and shipment-list/status views.
- PDF and supported image upload with file-type/signature checks and a 15 MB upload limit.
- PDF text extraction and LLM-assisted structured extraction.
- Field-level evidence and confidence values, with human review required.
- Cross-document validation for supported fields across invoice, packing list and BOL.
- Human corrections with an audit record of old value, new value, user and timestamp.
- Manager-only approval and manager-only Excel export, enforced by backend authorization and workflow gates.
- Export history and a multi-sheet Excel workbook.

Out of scope

- Direct customs portal filing or integration with ICEGATE or a production customs system.
- Official HSN classification, duty computation, legal advice or compliance certification.
- Production OCR for scanned/image-only PDFs.
- A verified integration with a customer's proprietary filing-software import template.

3. High-Level Architecture

Layer	Technology / responsibility
Frontend	React 19, Vite, React Router, Axios and Lucide icons. Provides dashboard, shipments, document upload, review and approval/export screens.
API	FastAPI. Exposes authentication, shipment, document, extraction, correction, validation, approval and export endpoints.
Persistence	SQLAlchemy ORM with SQLite as the default local database.
Document processing	PyMuPDF extracts embedded text from PDFs. Upload validation checks supported file types and signatures.
AI extraction	OpenAI Python SDK configured for the Groq-compatible API endpoint; model selected using GROQ_MODEL.
Security	bcrypt password hashing, signed JWT access tokens, role checks and environment-based secrets.
Export	openpyxl generates an Excel workbook and protects cells beginning with formula-trigger characters.

Logical flow: React browser client → FastAPI endpoints → SQLAlchemy/SQLite. Uploaded documents are processed by the PDF text extraction layer and AI extraction service. Extracted records are stored and sent through cross-document validation. The review UI presents evidence and discrepancies. The export service creates a workbook only after the backend approval gates pass.

4. Data Model

Entity	Important attributes	Purpose
User	id, name, email, password_hash, role, created_at	Identity and executive/manager authorization.
Shipment	id, shipment_number, importer, supplier, origin_country, status, created_at	Shipment-level business record.
Document	id, shipment_id, document_type, original_filename, stored_filename, extracted_data, uploaded_at	Uploaded source file and extracted structured data.
FieldCorrection	shipment_id, field_name, old_value, new_value, corrected_by, reason, corrected_at	Audit trail for human changes.
ExportHistory	shipment_id, filename, exported_by, export_status, exported_at	Tracks generated export attempts and outcomes.

A shipment has multiple documents. Documents belong to a shipment. Corrections and export-history entries are associated with a shipment.

5. Main Workflow

1. Authenticate: user signs in and receives a time-limited JWT. Passwords are stored as bcrypt hashes.
2. Create or select shipment: the user creates a shipment record or opens an existing shipment.
3. Upload documents: upload invoice, packing list and BOL. The API validates file extension, MIME/type expectations, signature and size before storing the file.
4. Extract: PDF text is extracted and sent to the configured LLM with a schema-oriented prompt. The response is parsed into structured fields with evidence and confidence where available.
5. Review: the interface displays extracted fields and source context. Review is required; low-confidence or missing values must be checked by a person.
6. Correct: an authorized user edits an incorrect value. The old and new values, editor and timestamp are recorded.

7. Validate: the backend compares supported fields across documents and reports CLEAR, REVIEW_REQUIRED or pending outcomes.
8. Approve: only a manager can approve, and approval is blocked while validation is not CLEAR or required human review remains incomplete.
9. Export: only a manager can generate the Excel workbook after approval and required-document checks pass. The export attempt is recorded.

6. AI Extraction Design

The extraction service uses the OpenAI SDK with a Groq-compatible base URL. GROQ_API_KEY and GROQ_MODEL are loaded from the backend environment. The prompt requests structured JSON, source evidence and confidence, and instructs the model not to invent values. Missing or unreadable values should remain unset rather than being guessed.

Supported examples include invoice/BOL identifiers and dates, importer/exporter/supplier details, currency, Incoterm, line-item descriptions, HSN values as printed in the source, quantities, units, unit prices, line totals, package counts/types, gross/net weights and container details where present. The system records extracted values for review; model output is not treated as authoritative.

All extracted data is subject to human confirmation.

Confidence and evidence are review aids, not calibrated statistical guarantees.

The invoice grand total is distinct from the sum of any displayed sample line items; the application must not silently conflate them.

Current PDF processing reads embedded text. Scanned PDFs require OCR and are a known limitation.

7. Cross-Document Validation and Workflow Gates

The validation layer compares supported fields between the uploaded documents and records matches, mismatches and pending checks. A mismatch changes the shipment to a review-required state. The package-type rule used in the demo requires the invoice and packing list to contain and agree on package type; a missing package type in the BOL does not by itself fail that particular comparison when the invoice and packing list agree.

Manager approval is rejected unless validation is CLEAR and required low-confidence fields have been human-verified.

Excel export is rejected unless manager approval exists, validation is CLEAR and all three required document types are present.

Role checks are performed by the API, not only by hiding frontend controls.

8. User Interface Decisions

Dashboard summarizes shipment activity and status.

Shipment list provides shipment identifiers, importer/supplier context, origin, status and actions.

Document workspace groups source files and extracted information by shipment.

Human Review presents extracted values for verification and correction.

Validation results expose matched, mismatched and pending checks.

Manager approval and export actions are separated from executive review actions.

9. Security and Configuration

Secrets are read from backend environment variables; .env is excluded from Git.

backend/.env.example contains placeholder values only.

Passwords are bcrypt-hashed; JWTs are signed with SECRET_KEY and expire after eight hours.

Backend endpoints enforce executive/manager permissions for restricted operations.

Upload size/type checks and Excel formula-injection protection reduce common input risks.

CORS is configured for the local Vite development origins (ports 5173 and 5174).

The current frontend API client uses the local backend URL http://127.0.0.1:8000. This is suitable for the demonstrated local setup, not a remote production deployment. A production deployment should configure an environment-driven API URL, HTTPS, managed database and file storage, secret management, monitoring, backups and a reviewed access-control policy.

10. Verification and Demonstration Results

Scenario	Observed result
Positive shipment SHP-24814	Invoice, packing list and BOL uploaded. Validation reported CLEAR with 10 matches, 0 mismatches and 0 pending checks. Manager approval succeeded and Excel export completed.
Negative shipment SHP-24815	BOL gross weight 3,165 kg differed from invoice/packing-list gross weight 3,120 kg. Validation reported REVIEW_REQUIRED with 9 matches and 1 mismatch. Approval and export were blocked with HTTP 409.
Executive attempts approval	Backend returned HTTP 403: only managers can approve shipments.
Executive attempts export	Backend returned HTTP 403: only managers can generate shipment exports.
Frontend production build	Vite production build completed successfully during local verification.
Excel generation dependency	openpyxl installed and import verified; demo workbook generation succeeded.

The positive and negative scenarios use synthetic demonstration documents and records. These are workflow tests, not evidence of production accuracy or legal validity.

11. Assumptions, Metrics and Limitations

Business case assumptions supplied for the exercise:

- 1,100 shipments per month; 45 minutes of documentation work per shipment; ₹350 per staff-hour.
- 3% error/amendment incidence and an assumed ₹9,000 cost per amendment or delay.
- The resulting annualized cost estimate is approximately ₹70.29 lakh under these assumptions. It is an illustrative estimate, not audited savings or a measured outcome.

Proposed success targets, not yet demonstrated as achieved:

- Reduce average documentation entry time to under 15 minutes.
- Reach at least 95% field-level accuracy measured over 200 labelled shipments.

- Reduce amended entries by 50% after three months of operational use.

Known limitations:

- No OCR for scanned PDFs; text-based PDFs are required for the current extraction path.
- No real customs portal, filing-software or external enterprise-system integration.
- No official HSN/duty determination; printed HSN values are extracted for human review only.
- SQLite and local file storage are development/demo choices, not a production scaling architecture.
- LLM output can be incomplete or incorrect and must be verified by a human.
- The generated Excel workbook is a demonstration layout, not a certified or customer-specific filing template.
- No 200-shipment labelled accuracy evaluation has been completed.

12. Future Improvements

Add OCR for scanned PDFs and image documents, with page-level source coordinates.

Create a labelled evaluation set and report precision/recall or field-level exact-match accuracy by document type and supplier.

Add configurable customer-specific validation rules and filing-software export templates.

Move database and documents to managed production services with encryption, retention policies and backups.

Use a frontend environment variable such as VITE_API_URL instead of a hardcoded local API address.

Add structured application logs, monitoring, rate limits, automated tests and CI/CD.

Integrate with approved external systems only after security, legal and operational review.

13. Demo Accounts

Role	Email	Password
Executive	executive@clearentry.ai	Executive@123
Manager	manager@clearentry.ai	Manager@123

These are seeded demo credentials and must be changed or disabled before any real deployment.